

# Ahmed Ikram | AI Engineer

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Final-year BSc Computer Science (Data Science & AI), Dundee (predicted First, June 2027). Built agentic-RAG, LLMops and ML systems in Agile delivery, including an NCR Atleos industry project with weekly stakeholder demos - translating progress for an engineer and a non-technical PM, shaping the build on feedback. Self-taught Terraform to production standard. Mentored junior pupils one-to-one. Seeking a graduate AI Engineering role from June 2027.

## EDUCATION

**University of Dundee** - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a First-Class Honours degree. Expected Graduation: June 2027

**High School of Dundee**

Sep 2018 - June 2023

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

## SKILLS

**Languages & Data:** Python · SQL · TypeScript · JavaScript · Bash · Rust / PyO3 · PostgreSQL · Redis · Kafka · Pydantic

**LLMs:** PyTorch · Hugging Face Transformers · QLoRA/LoRA/PEFT · AWQ · Unsloth · TRL · Accelerate · BitsandBytes · vLLM · Jinja2, Prompt & Context Engineering

**Agents & RAG:** LangGraph · LangChain · Model Context Protocol (MCP) · AWS Bedrock · ChromaDB · Cross-Encoder Reranking · Multimodal (Vision-Language) Extraction · Guardrails

**Evaluation & Observability:** RAGAS · LLM-as-Judge · LangSmith · Langfuse · Evidently AI · OpenTelemetry (Jaeger)

**MLOps & Serving:** MLflow · Weights & Biases · Apache Airflow · FastAPI · GraphQL · Modal · AWS SageMaker

**Classic ML:** XGBoost · Isolation Forest · scikit-learn · Optuna · Tesseract · OpenCV

**Cloud:** AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · Google Cloud (GCS) · Terraform · Docker · GitHub Actions / CI/CD · Linux

## PROJECTS

### ATM Log Aggregation, Anomaly Detection & Diagnostics - RAG over Logs and ML Anomaly Detection (NCR Atleos)

LangGraph · LangChain · MCP · ChromaDB · FastAPI · XGBoost · MLflow · Kafka · Redis · Docker · Terraform · AWS

- Built a Hybrid RAG as a LangGraph agent over a 12-tool MCP server with semantic chunking & cross-encoder reranking, grounding-gated re-retrieval and 4-signal confidence for refusal and citation (hallucination control); with OpenTelemetry GenAI spans to track query latency, token use, and gate decisions.
- Streamed agent responses via an SSE endpoint: reasoning blocks and tool-call steps as distinct events before the final response - backed by tiered conversation memory (Redis hot cache, Postgres cold archive) with turn-budget-gated compaction, degrading gracefully to hold token budget flat.
- Evaluated 3 retrieval strategies (baseline, hybrid, agentic) with RAGAS on a 50-query golden set: adaptive routing with re-retrieval doubles context recall (0.708 vs 0.350) with the best precision (0.874); a CI gate blocks RAGAS regression >0.05.
- Architected three-layer anomaly detection (XGBoost + Isolation Forest, Z-score, 7-heuristics) on a live Kafka Raft stream - 0.70 F1 at 97.3% precision via a 200-point Youden sweep - versioned in MLflow with champion aliases and cross-checked against a SageMaker endpoint.
- Terraform-provisioned AWS (ECS Fargate, RDS, S3, SageMaker, Secrets Manager, 4 least-privilege IAM roles, state locked in DynamoDB) behind OIDC CI/CD, validated by 1,887 tests across pytest, Vitest, Playwright and Terraform assertions.

### StockLens FinTech Spending & Investment App - AI Agent, Multimodal Extraction, MLOps

LangGraph · MCP · PyTorch · Optuna · Rust / PyO3 · Airflow · MLflow · Evidently AI · Tesseract · GraphQL · Redis · Terraform

- Built a 2-node ReAct StateGraph agent exposing 16 tools to multi-provider (AWS Bedrock) via a Stateless MCP server (Streamable HTTP, OAuth 2.1 PKCE). Correctness judged in CI by an LLM-as-judge with LangSmith tracing.
- Shipped a confidence-gated receipt-extraction cascade (Tesseract OCR → Bedrock Vision-LLM fallback) that only spends LLM budget when OCR confidence drops, matched to merchant with rapidfuzz.
- Trained a global LSTM (entity embeddings, Optuna 30-trial HPO, focal loss) reaching 51.6% five-day directional accuracy vs a 33% 3-class baseline (0.75 simulated Sharpe). 13 features computed in a Rust/PyO3 engine; retrained weekly via Airflow.

- Promotion requires beating the live champion by >2pp, gated by a binomial significance test ( $p < 0.05$ ), with Evidently drift monitoring (PSI/KS/JSD) before any model reaches production.
- Shipped on a fully Terraform-provisioned AWS estate (ECS Fargate, EFS champion serving) through 9 parallel CI jobs and an 8-job, 5-stage OIDC CD with 2,427 tests at a 90% coverage gate.

### **SWE-Qwen Automated LLM Fine-Tuning, Evaluation & Serving**

*Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · Terraform · Google Cloud*

- Architected a model-agnostic LLM Ops platform: one CLI command onboards any open-weight model into the pipeline - schema-validated data prep → QLoRA fine-tuning → execution-based evaluation → statistically-gated promotion → OpenAI-compatible serving; versioned GCS/W&B artifacts, with per-generation traces in Langfuse.
- Engineered a six-stage leak-free data pipeline: 20,477 raw SWE-bench issues → 17,456-examples across 46 repos; each repo held in exactly one split.
- Evaluation against the ground truth: every task materialises its repo in the official Docker image and executes the genuine pass/fail suite.
- Fine-tuned 3 QLoRA Qwen3-14B variants (4-bit NF4, Unsloth, FlashAttention on a Modal A100); champion hit 17.20% fail-to-pass on a 100-instance golden sample ( $7\times$  base, 95% CI 11.1-25.8%, McNemar  $p \approx 6e-05$  and 90.10% unrelated pass-pass rate, +61.6 pt vs base); promoted only past a 4-condition statistical gate.
- Served on Modal scale-to-zero with vLLM & per-request LoRA swapping and AWQ-quantisation or full-precision; a 20-instance F2P smoke gate blocks regressing PRs. GCP via Terraform and Workload Identity Federation; 1,539 tests.

### **CERTIFICATES**

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|----------------------------------------------------------------------------|-----------------|
| <u>AWS Academy Graduate - Machine Learning Foundations</u>                 | <i>Feb 2025</i> |
| <u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u> | <i>Oct 2024</i> |
| <u>AWS Academy Graduate - Cloud Foundations</u>                            | <i>Mar 2024</i> |

### **EXPERIENCE**

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|-------------------------------------------------|-----------------------------|
| <b>Beaully Mini Market</b> , Dundee, Scotland   |                             |
| <i>Retail Assistant</i>                         | <i>Sep 2023 - Present</i>   |
| <b>Zulus Peri Peri Grill</b> , Dundee, Scotland |                             |
| <i>Front of House Worker</i>                    | <i>June 2021 - Sep 2023</i> |